

Alpa

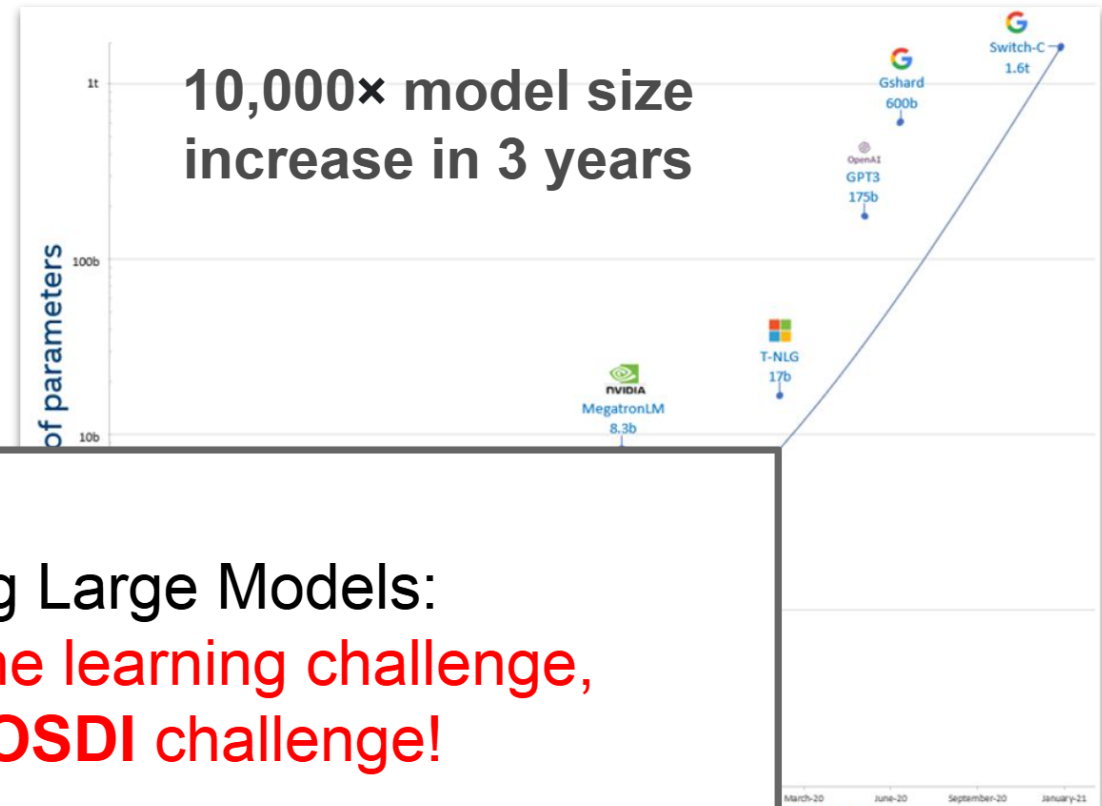
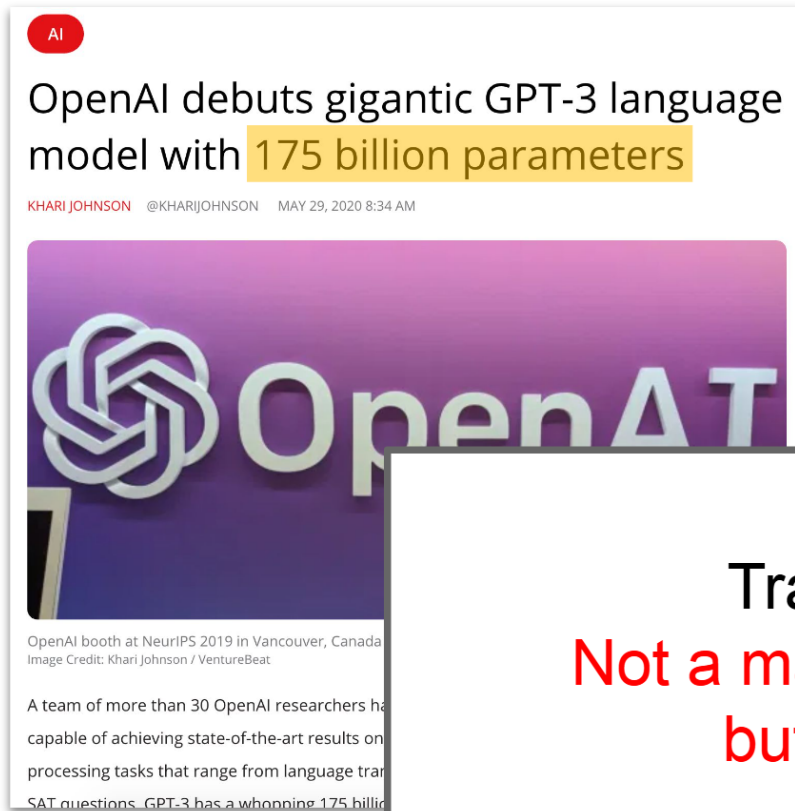
Automating Inter- and Intra-Operator
Parallelism for Distributed Deep Learning

2025.01.14 이유찬

논문 정보

- ▶ 16th USENIX Symposium on Operating Systems Design and Implementation (USENIX OSDI 22)
- ▶ Lianmin Zheng, Zhuohan Li, and Hao Zhang, et al
- ▶ UC Berkeley, AWS, Google, Shanghai Jiao Tong Univ, Carnegie Mellon Univ, Duke Univ

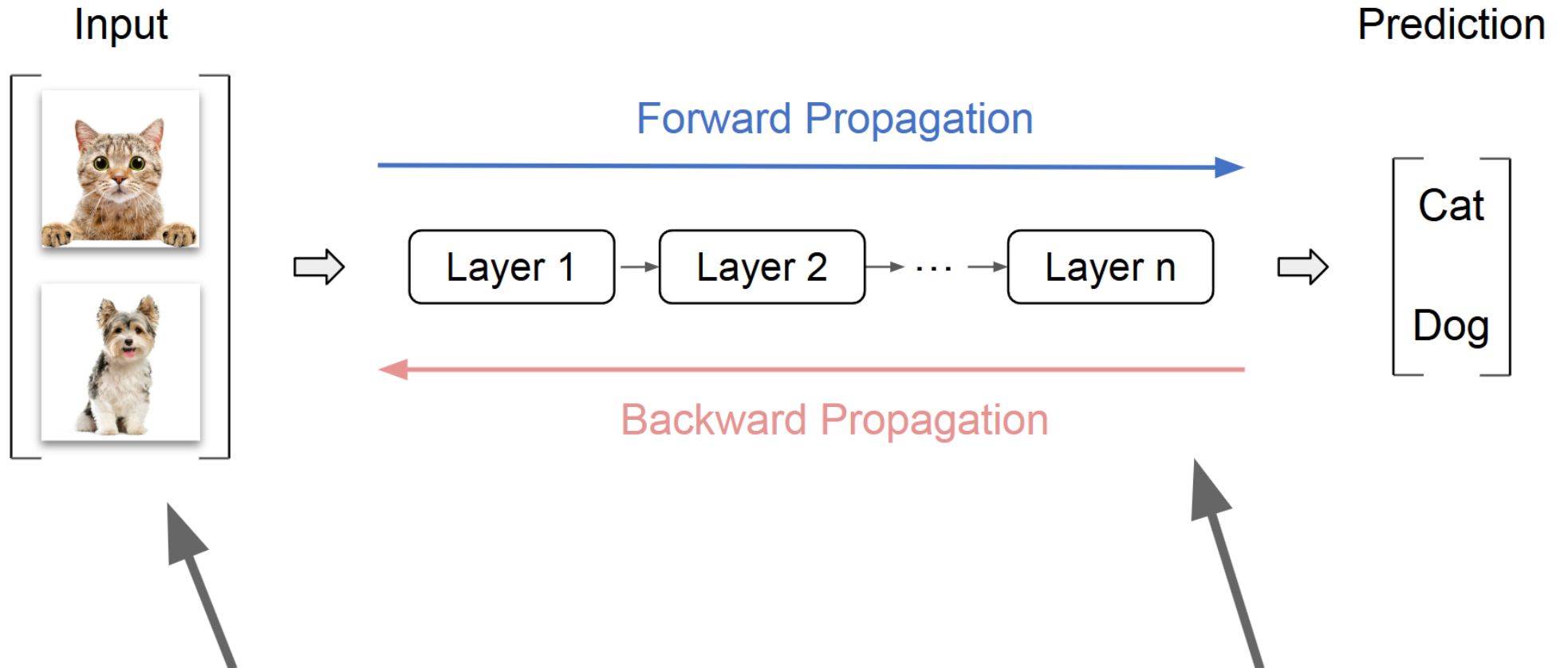
Large Models Enable Breakthroughs in Machine Learning



Training Large Models:
Not a machine learning challenge,
but an **OSDI** challenge!

Image source: towardsdatascience.com

Challenges

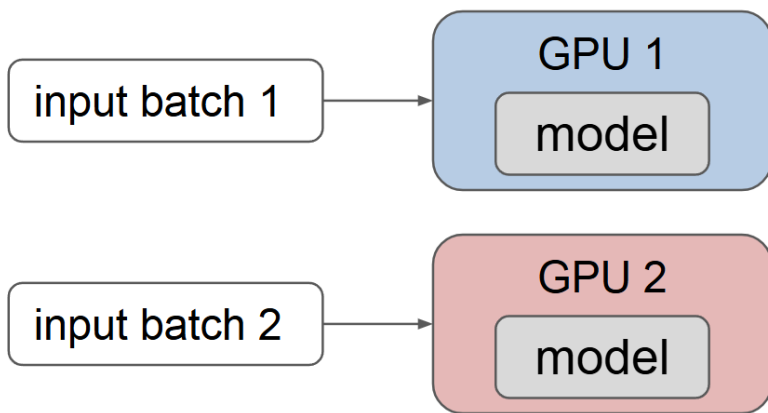


1. What if the **input dataset** is very large?

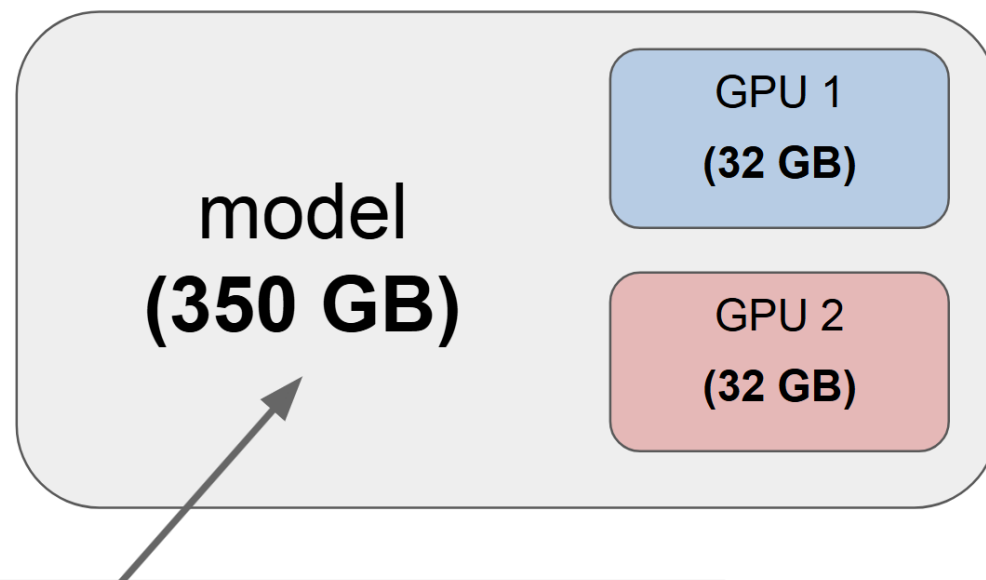
2. What if the **model** is very large?

Challenges

- ▶ Data가 큰 경우
 - ▶ 비교적 쉬움
 - ▶ 모델을 복제하고 배치를 나눔

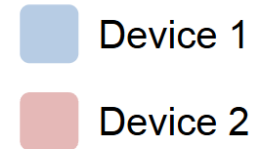
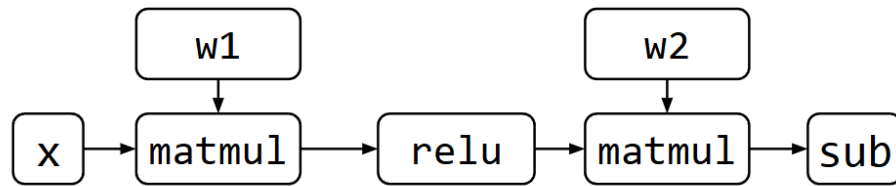


- ▶ 모델이 큰 경우
 - ▶ 비교적 어려움

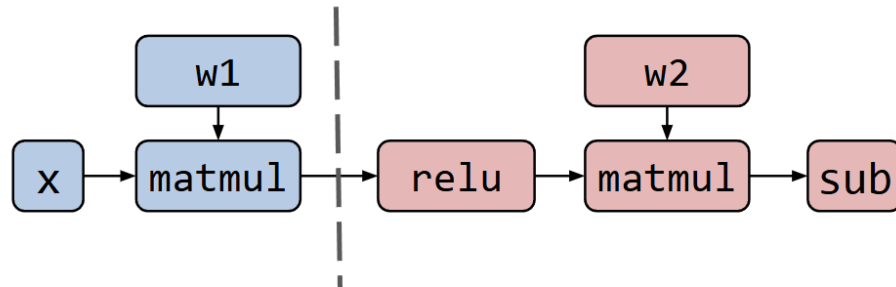


Challenge: How to partition a computational graph?

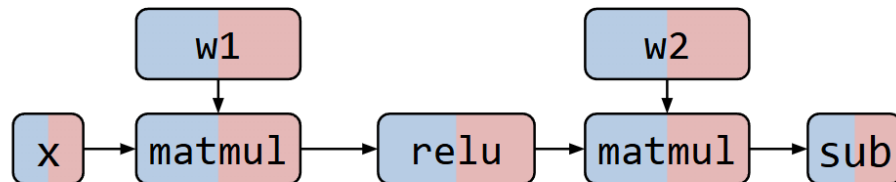
Partition Computational Graphs



Strategy 1: Inter-operator Parallelism



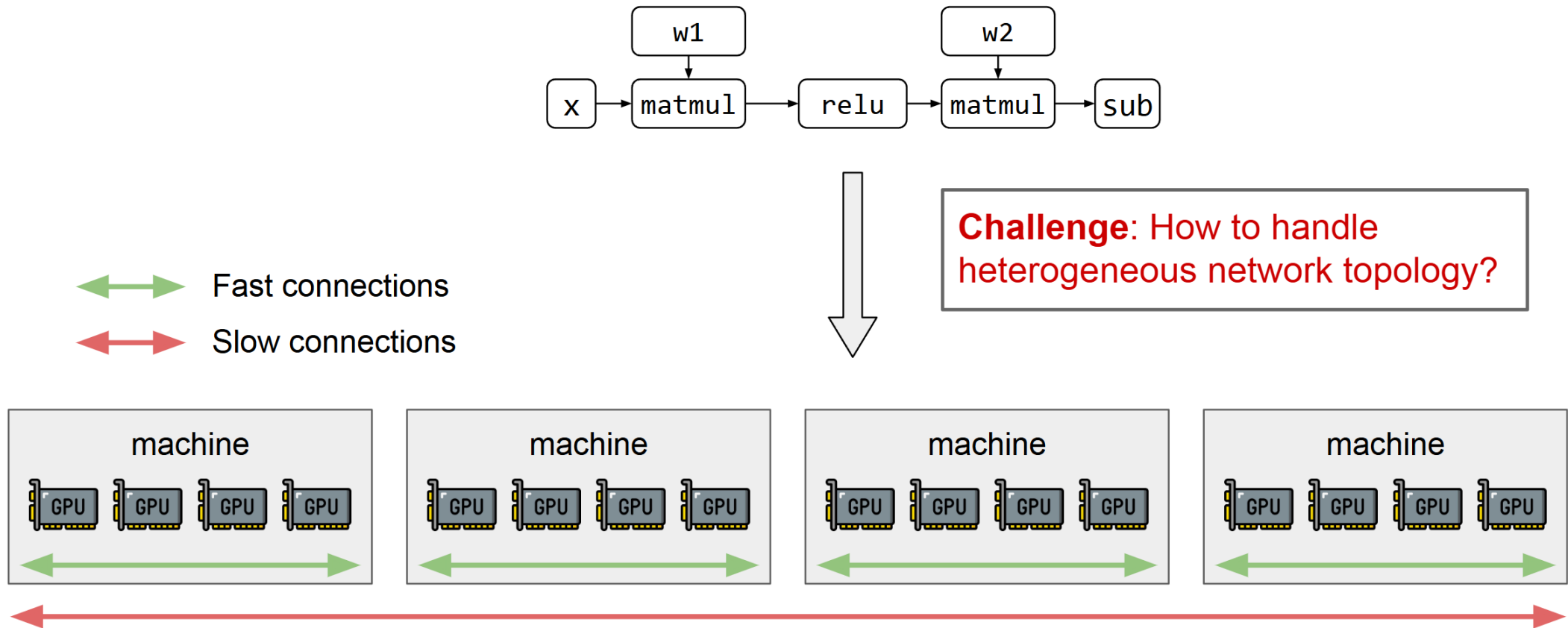
Strategy 2: Intra-operator Parallelism



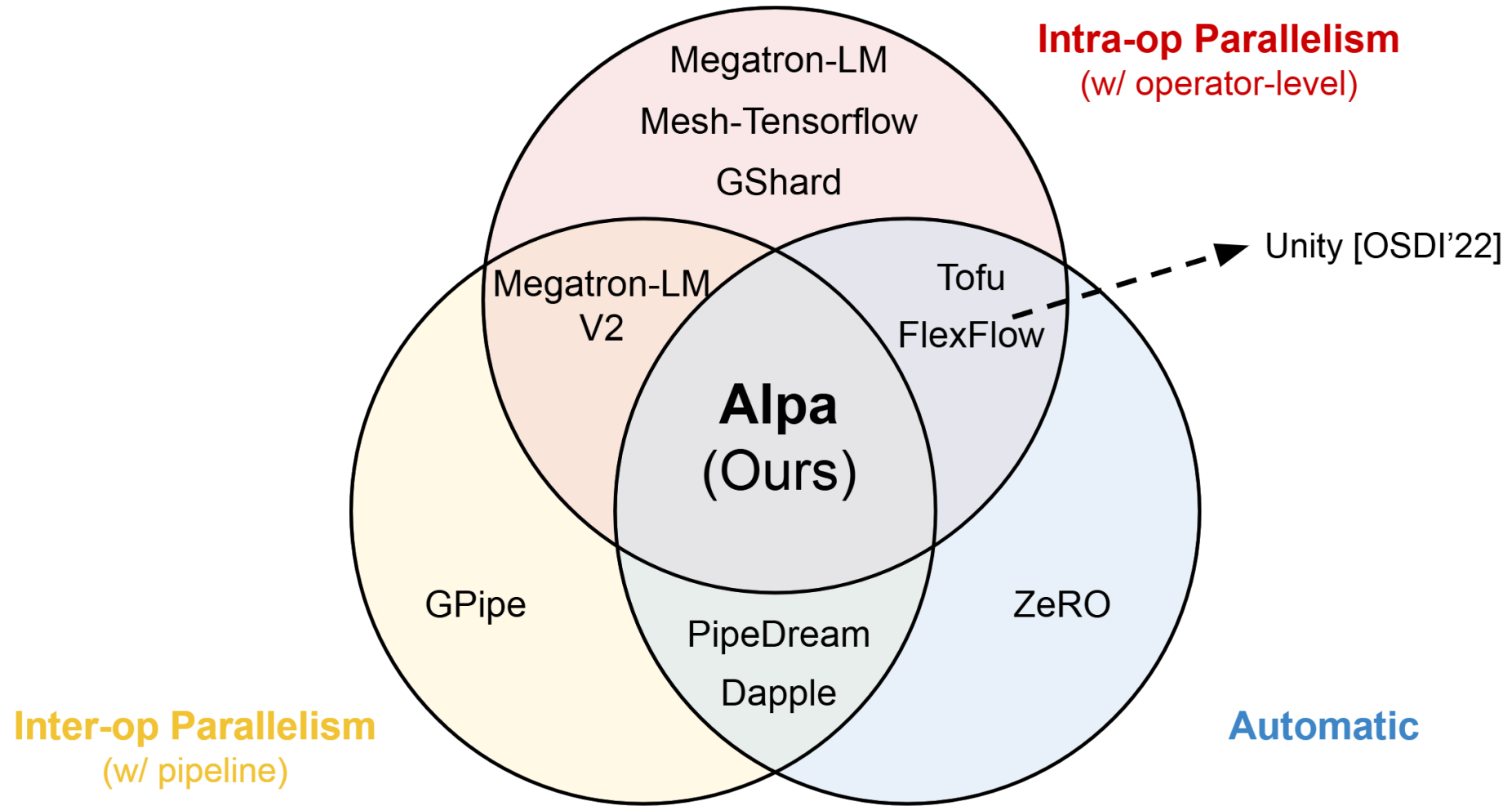
Trade-off

	Inter-operator Parallelism	Intra-operator Parallelism
Communication	Less	More
Device Idle Time	More	Less

Heterogeneity



Prior Works



Alpa Compiler

Contribution



Two-level hierarchical space of parallelism techniques.



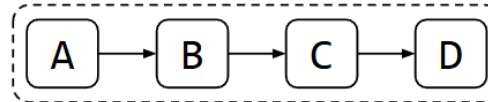
Effective optimization algorithms at each level.



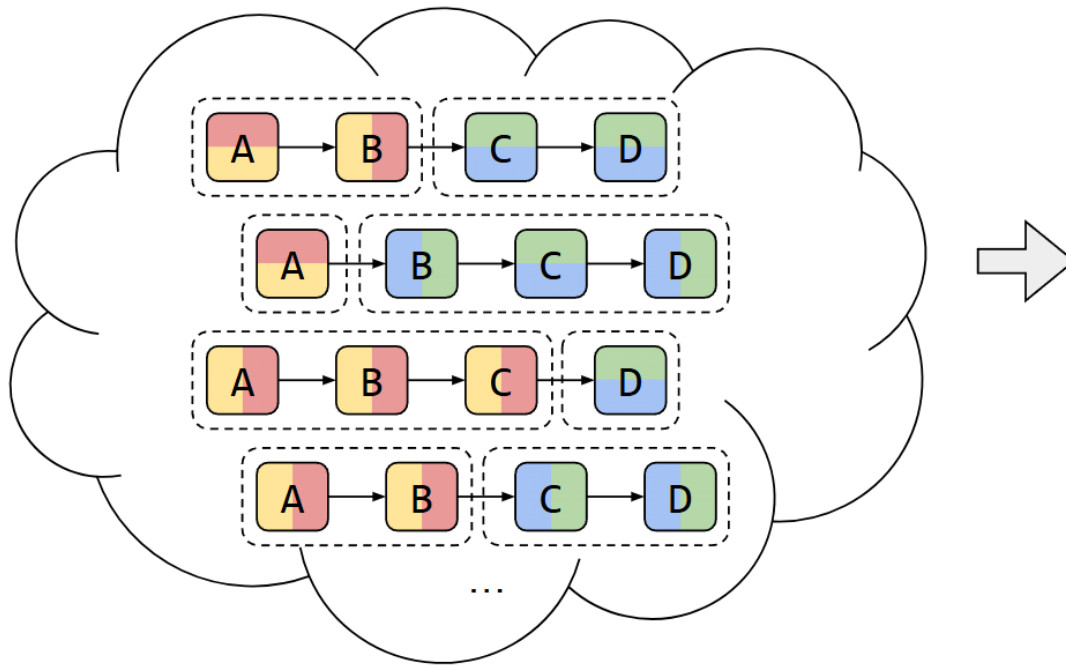
Efficient compiler and runtime system implementation.

Optimization

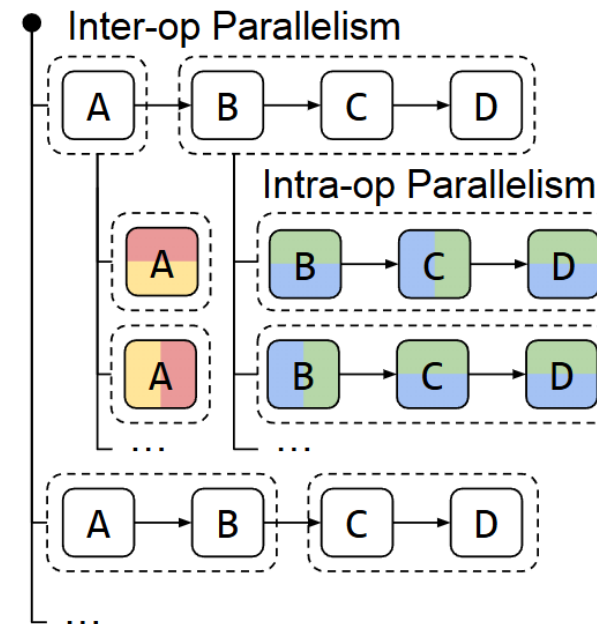
Computational Graph



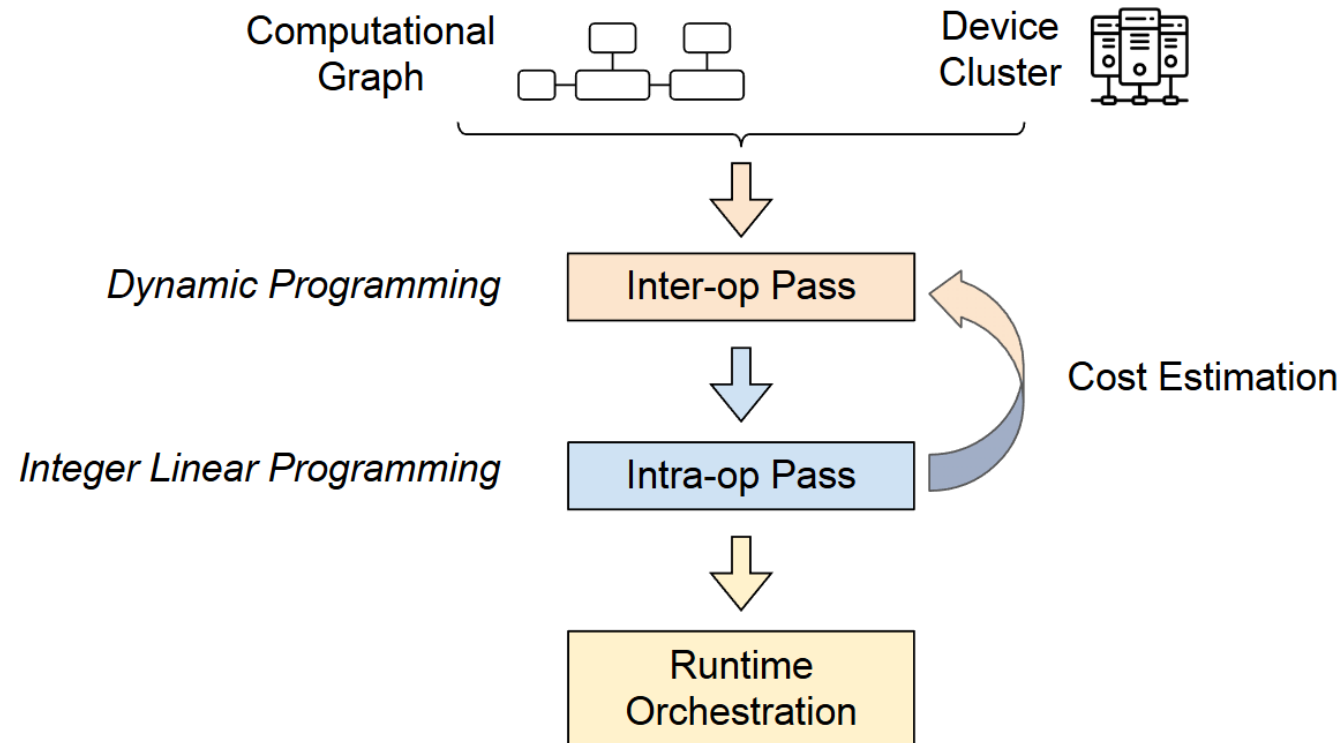
Whole Search Space



Alpha Hierarchical Space

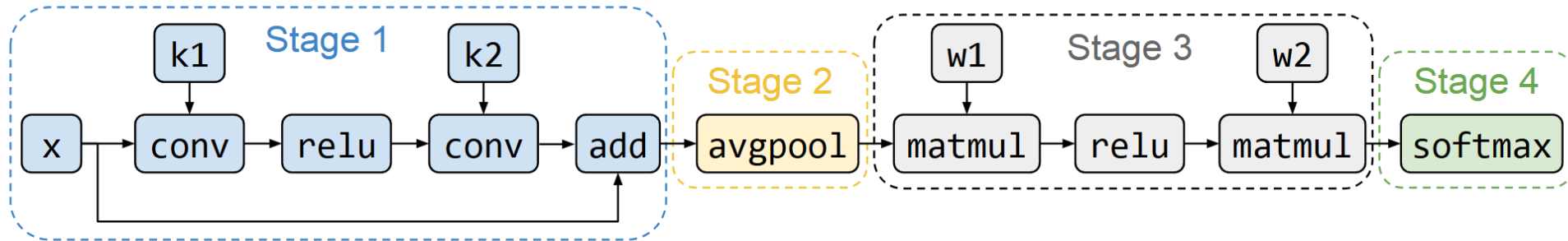


Optimization

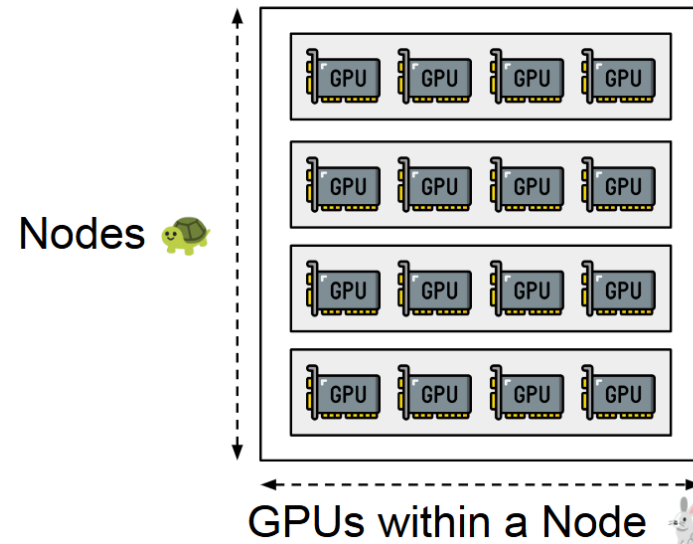


Inter-op optimization

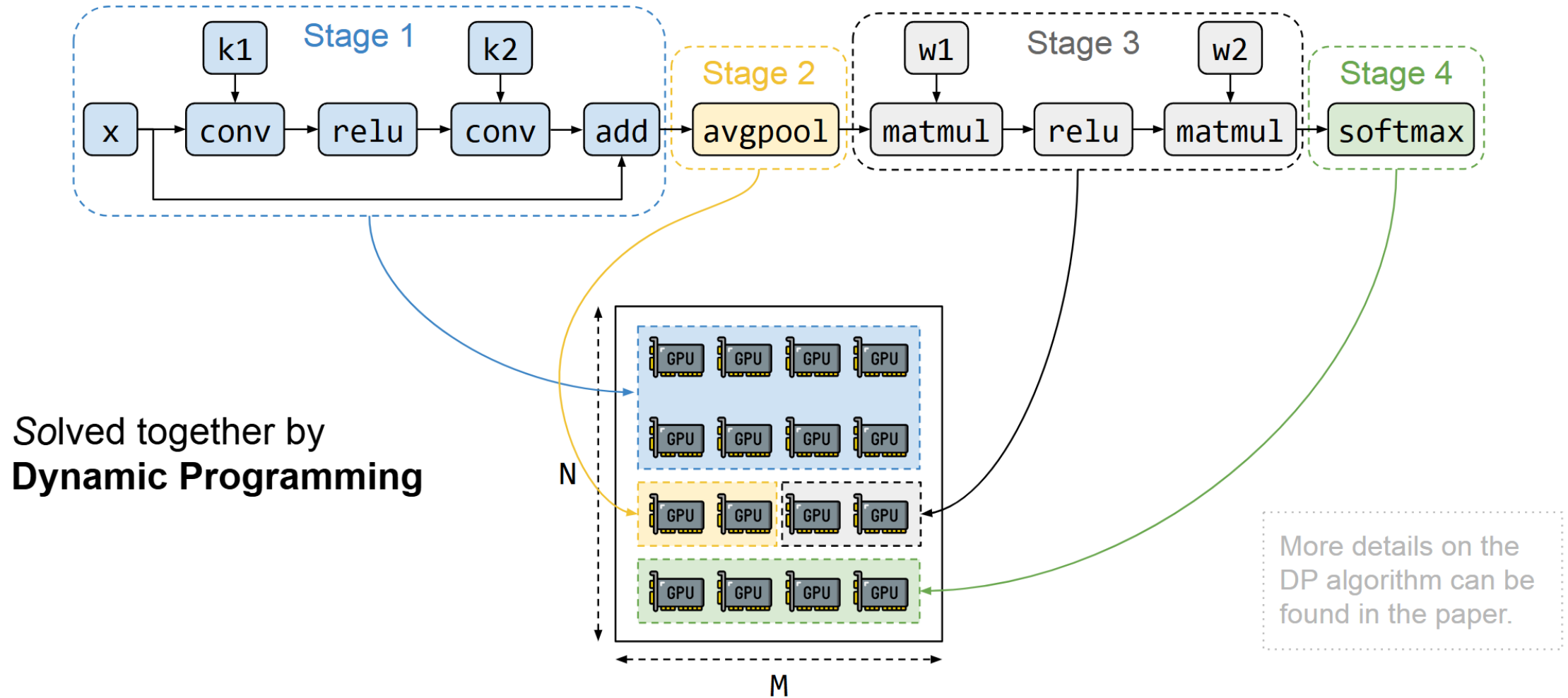
Partitioned Computational Graph



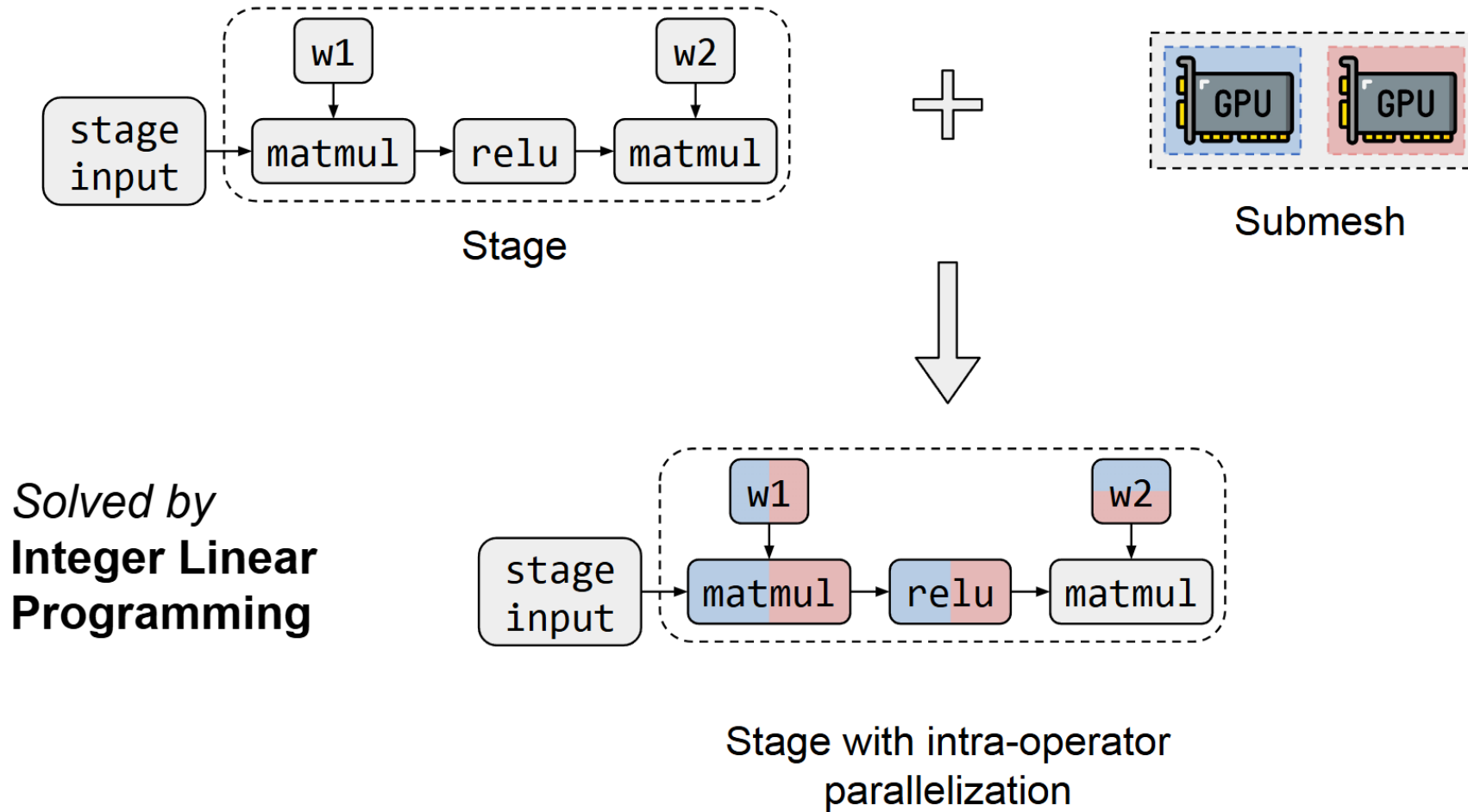
Cluster (2D Device Mesh)



Inter-op optimization



Intra-op optimization



Intra-Op parallelism

► Sharding spec

Table 1: Sharding specs of a 2-dimensional tensor on a 2×2 device mesh. A is a (N, M) tensor. The device mesh is [[Device 0, Device 1], [Device 2, Device 3]]. Each device stores a partition of A . The first column is the name of the sharding spec. The latter columns use Numpy syntax to describe the partitions stored on each device.

Spec	Device 0	Device 1	Device 2	Device 3
RR	$A[0:N, 0:M]$	$A[0:N, 0:M]$	$A[0:N, 0:M]$	$A[0:N, 0:M]$
S^0S^1	$A[0:\frac{N}{2}, 0:\frac{M}{2}]$	$A[0:\frac{N}{2}, \frac{M}{2}:M]$	$A[\frac{N}{2}:N, 0:\frac{M}{2}]$	$A[\frac{N}{2}:N, \frac{M}{2}:M]$
S^1S^0	$A[0:\frac{N}{2}, 0:\frac{M}{2}]$	$A[\frac{N}{2}:N, 0:\frac{M}{2}]$	$A[0:\frac{N}{2}, \frac{M}{2}:M]$	$A[\frac{N}{2}:N, \frac{M}{2}:M]$
S^0R	$A[0:\frac{N}{2}, 0:M]$	$A[0:\frac{N}{2}, 0:M]$	$A[\frac{N}{2}:N, 0:M]$	$A[\frac{N}{2}:N, 0:M]$
S^1R	$A[0:\frac{N}{2}, 0:M]$	$A[\frac{N}{2}:N, 0:M]$	$A[0:\frac{N}{2}, 0:M]$	$A[\frac{N}{2}:N, 0:M]$
RS^0	$A[0:N, 0:\frac{M}{2}]$	$A[0:N, 0:\frac{M}{2}]$	$A[0:N, \frac{M}{2}:M]$	$A[0:N, \frac{M}{2}:M]$
RS^1	$A[0:N, 0:\frac{M}{2}]$	$A[0:N, \frac{M}{2}:M]$	$A[0:N, 0:\frac{M}{2}]$	$A[0:N, \frac{M}{2}:M]$
$S^{01}R$	$A[0:\frac{N}{4}, 0:M]$	$A[\frac{N}{4}:\frac{N}{2}, 0:M]$	$A[\frac{N}{2}:\frac{3N}{4}, 0:M]$	$A[\frac{3N}{4}:N, 0:M]$
RS^{01}	$A[0:N, 0:\frac{M}{4}]$	$A[0:N, \frac{M}{4}:\frac{M}{2}]$	$A[0:N, \frac{M}{2}:\frac{3M}{4}]$	$A[0:N, \frac{3M}{4}:M]$

► Resharding spec

Table 2: Several cases of resharding. $all-gather(x, i)$ means an all-gather of x bytes along the i -th mesh axis. M is the size of the tensor. (n_0, n_1) is the mesh shape.

#	Src Spec	Dst Spec	Communication Cost
1	RR	S^0S^1	0
2	S^0R	RR	$all-gather(M, 0)$
3	S^0S^1	S^0R	$all-gather(\frac{M}{n_0}, 1)$
4	S^0R	RS^0	$all-to-all(\frac{M}{n_0}, 0)$
5	S^0S^1	$S^{01}R$	$all-to-all(\frac{M}{n_0 \cdot n_1}, 1)$

Intra-Op parallelism

► Parallel algorithms

Table 3: Several parallel algorithms for a batched matmul $C_{b,i,j} = \sum_k A_{b,i,k} B_{b,k,j}$. The notation $all-reduce(x, i)$ means an all-reduce of x bytes along the i -th mesh axis. M is the size of the output tensor. (n_0, n_1) is the mesh shape.

#	Parallel Mapping	Output Spec	Input Specs	Communication Cost
1	$i \rightarrow 0, j \rightarrow 1$	RS^0S^1	RS^0R, RRS^1	0
2	$i \rightarrow 0, k \rightarrow 1$	RS^0R	RS^0S^1, RS^1R	$all-reduce(\frac{M}{n_0}, 1)$
3	$j \rightarrow 0, k \rightarrow 1$	RRS^0	RRS^1, RS^1S^0	$all-reduce(\frac{M}{n_0}, 1)$
4	$b \rightarrow 0, i \rightarrow 1$	S^0S^1R	S^0S^1R, S^0RR	0
5	$b \rightarrow 0, k \rightarrow 1$	S^0RR	S^0RS^1, S^0S^1R	$all-reduce(\frac{M}{n_0}, 1)$
6	$i \rightarrow \{0, 1\}$	$RS^{01}R$	$RS^{01}R, RRR$	0
7	$k \rightarrow \{0, 1\}$	RRR	$RRS^{01}, RS^{01}R$	$all-reduce(M, \{0, 1\})$

► ILP(Integer Linear Programming)

$$\min_s \sum_{v \in V} s_v^T (c_v + d_v) + \sum_{(v,u) \in E} s_v^T R_{vu} s_u, \quad (1)$$

s: parallel algorithm

d: communication cost

c: compute cost

R: resharding cost

C는 0으로 set

Inter-Op parallelism

► Pipeline parallelism

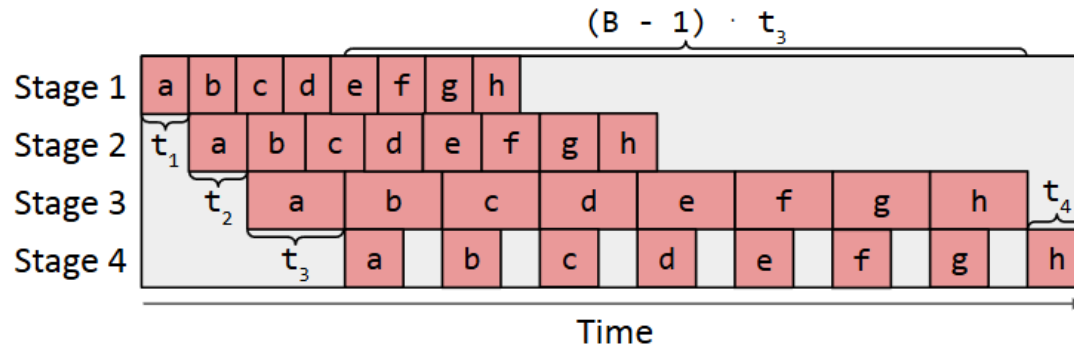


Figure 5: Illustration of the total latency of a pipeline, which is determined by two parts: the total latency of all stages ($t_1 + t_2 + t_3 + t_4$) and the latency of the slowest stage ($(B-1) \cdot t_3$).

► DP (Dynamic Programming)

$$T^* = \min_{\substack{s_1, \dots, s_S; \\ (n_1, m_1), \dots, (n_S, m_S)}} \left\{ \sum_{i=1}^S t_i + (B-1) \cdot \max_{1 \leq j \leq S} \{t_j\} \right\}.$$

$$F(s, k, d; t_{\max}) = \min_{\substack{k \leq i \leq K \\ n_s \cdot m_s \leq d}} \left\{ \begin{array}{l} t_{\text{intra}}((o_k, \dots, o_i), \text{Mesh}(n_s, m_s), s) \\ + F(s-1, i+1, d - n_s \cdot m_s; t_{\max}) \\ | t_{\text{intra}}((o_k, \dots, o_i), \text{Mesh}(n_s, m_s), s) \leq t_{\max} \end{array} \right\}, \quad (3)$$

$$G(k, r) = \min_{1 \leq i \leq k} \left\{ \begin{array}{l} \max\{G(i-1, r-1), C(i, k)\} \\ | FLOP(o_i, \dots, o_k) \leq \frac{(1+\delta)FLOP_{\text{total}}}{L} \end{array} \right\}, \quad (6)$$

Evaluation

Set up

- ▶ 8 nodes 64 GPUs
 - ▶ 8 NVIDIA V100 16GB
 - ▶ 64 vCPUs
 - ▶ 488GB memory
 - ▶ 노드 내 8 GPU는 NVLINK
 - ▶ 노드 간 연결은 25Gbps

Table 4: Models used in the end-to-end evaluation. LM = language model. IC = image classification.

Model	Task	Batch size	#params (billion)	Precision
GPT-3 [10]	LM	1024	0.35, 1.3, 2.6, 6.7, 15, 39	FP16
GShard MoE [31]	LM	1024	0.38, 1.3, 2.4, 10, 27, 70	FP16
Wide-ResNet [59]	IC	1536	0.25, 1.0, 2.0, 4.0, 6.7, 13	FP32

- ▶ 학습 처리량 비교 (PFLOPS)
- ▶ 모델 사이즈 scaling

Evaluation

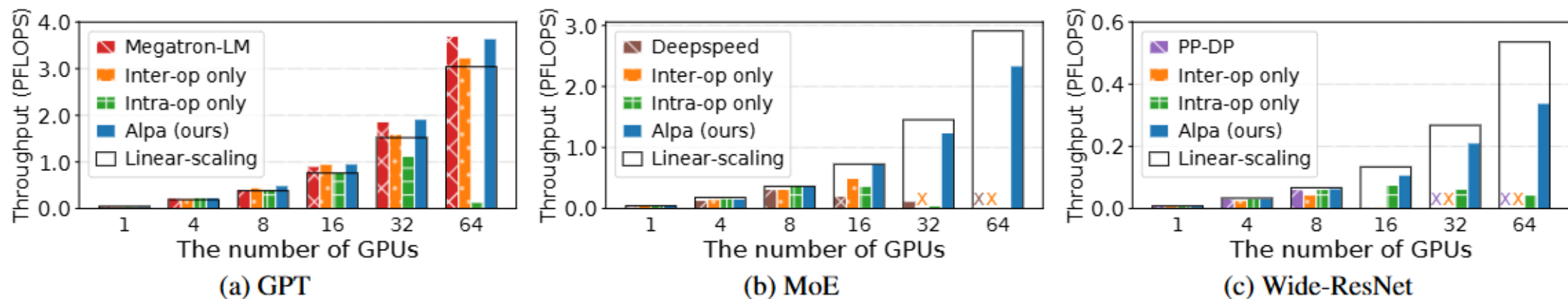


Figure 7: End-to-end evaluation results. “x” denotes out-of-memory. Black boxes represent linear scaling.

▶ GPT

- ▶ Megatron과 유사한 성능

▶ MoE

- ▶ Deepspeed는 inter-op만 사용

Evaluation

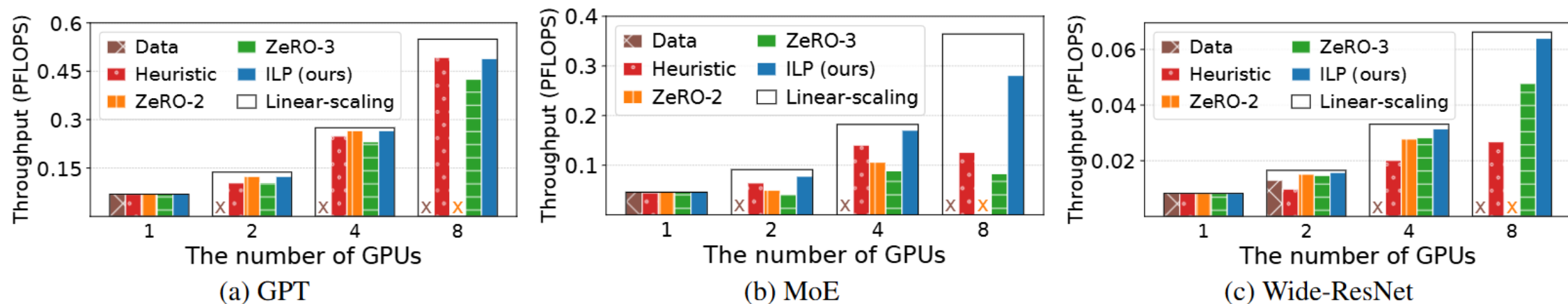


Figure 8: Intra-operator parallelism ablation study. “x” denotes out-of-memory. Black boxes represent linear scaling.